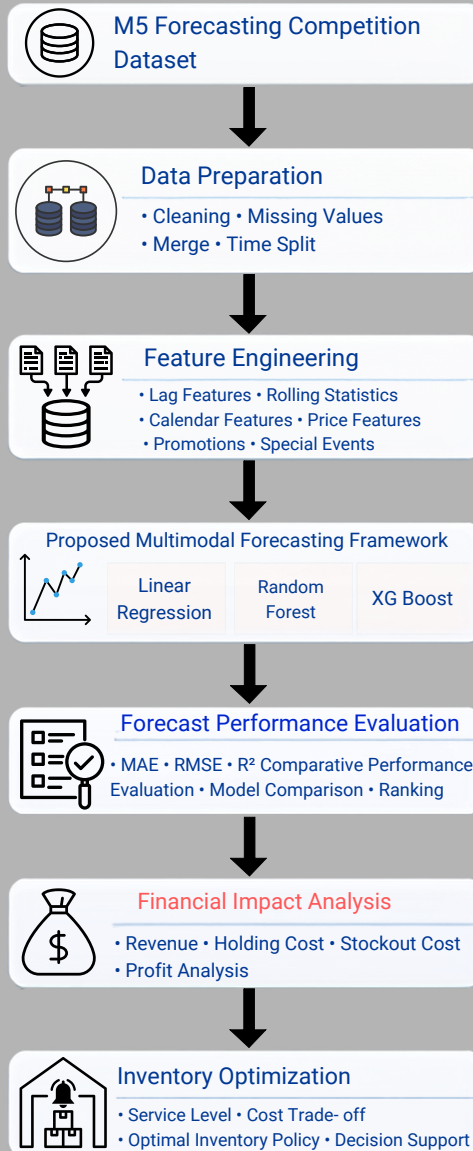


Graphical Abstract

A Data-Driven Framework for Retail Demand Forecasting and Financial Impact Evaluation Using the M5 Benchmark Dataset

Mohamed Meraaj Mansur Sharif, Raja Hashim Ali, Talha Ali Khan

Proposed Research Framework



External Validation using the DataCo Supply Chain Dataset

**Managerial Decision Support
&
Financial Decision Making**

Highlights

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- Reproducible forecasting links demand prediction with inventory decisions.
- M5 analysis uses the 500 highest-volume item–store series.
- XGBoost achieves MAE 5.582 and RMSE 8.690.
- Rolling demand features dominate the Random Forest importance diagnostic.
- Forecasts are translated into financial and safety-stock scenarios.

A Data-Driven Framework for Retail Demand Forecasting and Financial Impact Evaluation Using the M5 Benchmark Dataset

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Abstract

Retail demand forecasting is important for inventory planning, financial analysis, and supply-chain decision-making, but predictive accuracy alone does not show how forecast errors translate into operational consequences. This study develops a reproducible machine-learning framework that connects retail demand forecasting with financial and inventory scenario analysis using the M5 Forecasting dataset. The study selects the 500 highest-volume item-store series using training-period demand, integrates sales, calendar, and weekly price data, and engineers lag, rolling-window, pricing, and calendar-derived features. Linear Regression, Random Forest, and XGBoost are evaluated under identical conditions using a chronological 28-day hold-out containing 14,000 observations, while LightGBM and ARIMA are used as supporting analyses. The full feature configuration improves Random Forest MAE from 5.842 to 5.608, while the 7-day rolling mean accounts for 79.9% of Random Forest feature importance. Among the three core models, XGBoost achieves the strongest row-level performance with an MAE of 5.582, RMSE of 8.690, and R^2 of 0.705. Using LightGBM forecasts for downstream decision analysis, the illustrative scenario produces approximately \$335,984.96 in fulfilled-demand revenue, \$6,020.06 in stockout-cost exposure, and \$3,273.91 in holding-cost exposure. Estimated safety stock increases from approximately 12.0 units at a 70% service target to 37.8 units at 95%. External validation on the DataCo SMART SUPPLY CHAIN dataset produces an MAE of 50.32 and RMSE of 61.25 on its own scale. The framework therefore provides a transparent link between predictive performance, uncertainty, inventory requirements, and financial consequences while explicitly distinguishing empirical results from illustrative decision scenarios.

Keywords: Demand forecasting, Machine learning, Retail supply chain, Inventory optimisation, Financial impact analysis, M5 forecasting dataset, Decision support

1. Introduction

Retail demand forecasting sits near the centre of procurement, replenishment, workforce planning, and service-level decisions in modern supply chains [? ?]. Forecast errors have asymmetric operational consequences because under-forecasting can contribute to stockouts and lost sales, whereas over-forecasting can leave capital tied up in excess inventory and increase holding costs. The value of a forecasting system therefore depends not only on predictive accuracy but also on how its errors interact with inventory and cost decisions. Retailers increasingly have access to transactional, pricing, calendar, and product-level information that can be incorporated into machine-learning forecasting pipelines [? ?]. Tree-based ensemble methods are particularly attractive for structured retail data because they can model nonlinear relationships while accommodating heterogeneous engineered features [? ? ?]. At the same time, the forecasting literature continues to debate the relative value of statistical, machine-learning, and deep-learning approaches across datasets and forecast horizons [? ? ?]. The present study focuses on a transparent tabular machine-learning setting rather than attempting to claim universal superiority for any single model family. The empirical analysis uses the M5 Forecasting benchmark and evaluates the predictive and downstream implications of forecasts within one consistent experimental pipeline.

1.1. Gap Analysis

A recurring limitation in the retail forecasting literature is the separation between statistical forecast accuracy and the operational or financial value of those forecasts. Studies frequently report MAE, RMSE, or related accuracy measures without translating prediction errors into inventory exposure, revenue implications, or service-level requirements [? ?]. Conversely, studies that connect forecasts with inventory decisions often use different datasets, different forecasting procedures, or simulation settings that make direct comparison difficult. Recent work has strengthened probabilistic forecasting and forecast-driven inventory analysis, but the two evaluation layers

are still commonly treated as separate methodological exercises [? ?]. This separation matters because a small improvement in RMSE does not necessarily produce a proportional reduction in stockout or holding costs. The operational consequence of an error depends on demand level, forecast uncertainty, selling price, cost assumptions, lead time, and the inventory policy used to translate the prediction into action. There is therefore a need for transparent empirical frameworks in which model accuracy, uncertainty, and decision-oriented quantities are derived from the same held-out observations. This study addresses that gap by carrying a single M5-based pipeline from feature engineering and forecasting through financial and inventory scenario analysis.

1.2. Research Questions

The study investigates one main research question and six supporting questions:

1. **RQ1:** How accurately can machine-learning models forecast retail product demand using a multi-source feature set built from the M5 dataset, and how much of that accuracy comes from combining historical, pricing, and calendar information rather than from any single source alone?
2. **RQ2:** Does multi-source feature engineering improve forecasting performance relative to a narrower feature set, and which engineered feature families contribute the most predictive signal?
3. **RQ3:** Which of Linear Regression, Random Forest, and XGBoost provides the most reliable predictive performance under identical experimental conditions?
4. **RQ4:** How does forecast output translate into inventory-related financial performance, including revenue, stockout cost, and holding cost?
5. **RQ5:** Can demand forecasting support inventory optimisation by quantifying the trade-off between target service level and safety-stock investment?
6. **RQ6:** How does forecasting uncertainty relate to stockout risk, and what does that relationship imply for safety-stock policy?
7. **RQ7:** Does forecasting performance remain stable across low-, medium-, and high-demand regimes, and does the underlying feature-engineering approach generalise beyond the M5 benchmark?

67 *1.3. Problem Statement*

68 The problem addressed in this study is that retail organisations need
69 forecasting systems that are evaluated not only on statistical accuracy but
70 also on the financial and operational consequences of that accuracy. The
71 study addresses this problem for the M5 Forecasting dataset by engineer-
72 ing a multi-source feature set from historical sales, calendar, and pricing
73 data. Three forecasting algorithms of increasing complexity are trained un-
74 der identical inputs and a chronological validation design. The resulting
75 predictions are then translated into revenue, stockout cost, holding cost,
76 and service-level-dependent safety-stock requirements. Forecast uncertainty
77 is further examined by separating observations into uncertainty tiers and
78 relating error magnitude to stockout risk. Demand-regime analysis is used
79 to identify whether forecast performance is uniform across low-, medium-,
80 and high-demand series. An external DataCo experiment provides a sepa-
81 rate robustness check without treating the different target scale as directly
82 comparable with M5. The resulting framework therefore answers both how
83 accurately demand can be forecast and what that forecast accuracy means
84 for inventory-oriented decisions.

85 *1.4. Novelty of This Study*

86 The contribution is methodological rather than a claim of state-of-the-art
87 leaderboard performance. The study combines several evaluation layers that
88 are commonly reported separately and keeps the underlying observations and
89 validation horizon fixed.

- 90 • **Unified evaluation:** Linear Regression, Random Forest, and XG-
91 Boost are compared using the same preprocessing, feature matrix, chrono-
92 logical split, and test observations.
- 93 • **Decision translation:** The same held-out forecasting outputs are
94 translated into revenue, stockout-cost, holding-cost, and safety-stock
95 scenarios.
- 96 • **Uncertainty-aware analysis:** Forecast-error magnitude is used to
97 differentiate safety-stock requirements and identify higher-risk obser-
98 vations.
- 99 • **Regime analysis:** Forecast accuracy is evaluated separately across
100 low-, medium-, and high-demand regimes.

101 • **External validation:** The core lag/rolling feature-engineering strat-
102 egy is examined on the DataCo SMART SUPPLY CHAIN dataset
103 while explicitly avoiding direct metric equivalence across different tar-
104 get scales.

105 1.5. *Significance of Our Work*

106 The study provides a compact decision-support perspective on retail fore-
107 casting in which prediction quality and downstream consequences are evalu-
108 ated together without conflating them. The results show that recent demand
109 history carries most of the predictive signal in the selected high-volume M5
110 series. The strongest core model is XGBoost, but the performance difference
111 from Random Forest is small. The downstream analysis shows that fore-
112 cast uncertainty is unevenly distributed and that safety-stock requirements
113 increase substantially as the target service level becomes more demanding.
114 The financial quantities are deliberately treated as scenario outputs because
115 retailer-specific procurement, holding, shortage, and lost-sales costs are un-
116 available in the public benchmark. This distinction prevents the dollar values
117 from being interpreted as observed Walmart financial performance. The ex-
118 ternal validation further indicates that the basic temporal feature-engineering
119 strategy can be transferred to a structurally different supply-chain dataset,
120 although the experiment is not a controlled cross-dataset benchmark. To-
121 gether, these findings support a transparent forecasting framework in which
122 model accuracy is considered as one component of a broader inventory deci-
123 sion problem.

124 2. Literature Review

125 Retail demand forecasting spans statistical time-series models, tree-based
126 machine learning, deep-learning sequence models, feature-engineering ap-
127 proaches, and decision-oriented inventory analytics. Classical statistical meth-
128 ods such as ARIMA remain attractive because of their transparency and low
129 computational requirements, while machine-learning models can incorporate
130 external variables such as price and calendar information [? ?]. Tree-based
131 ensembles have performed strongly on structured retail datasets, including
132 the M5 competition [? ? ?]. Deep-learning models such as LSTM and
133 Transformer architectures provide alternative mechanisms for learning tem-
134 poral dependencies and probabilistic representations [? ? ?]. A parallel

135 literature emphasises feature engineering and the value of multi-source pre-
 136 dictors, particularly lag and rolling-window statistics [? ? ?]. Finally, in-
 137 ventory research increasingly examines how forecasts can be translated into
 138 stocking decisions, with recent work emphasising the non-linear relationship
 139 between forecast accuracy and inventory performance [? ? ?]. The following
 140 subsections position the present study within these strands.

141 *2.1. Tree-Based Ensemble Learning*

142 Random Forest uses bootstrap aggregation and random feature selection
 143 to reduce the variance of decision-tree predictors [? ?]. Its suitability for
 144 tabular forecasting arises from its ability to represent nonlinear interactions
 145 without requiring extensive functional assumptions. XGBoost extends the
 146 tree-ensemble family through sequential gradient boosting and regularisation
 147 [? ?]. The M5 competition demonstrated the effectiveness of carefully
 148 engineered boosted-tree systems at large retail forecasting scale [? ?].
 149 More recent retail studies have continued to report strong performance from
 150 simplified or tuned tree-based methods, suggesting that feature engineering
 151 and implementation discipline can matter as much as model complexity [?
 152 ? ?]. The present study therefore compares Linear Regression, Random
 153 Forest, and XGBoost directly rather than assuming that the most complex
 154 model must perform best. LightGBM is included only as a supporting model
 155 for diagnostics and downstream scenario analysis. This separation keeps the
 156 headline comparison interpretable while allowing a closely related boosting
 157 model to support the decision-oriented experiments.

158 *2.2. Deep Learning and Hybrid Forecasting*

159 LSTM networks model sequential dependencies through recurrent mem-
 160 ory mechanisms and have been widely applied to demand forecasting [? ?
 161]. Transformer architectures use attention mechanisms to represent long-
 162 range dependencies and have become increasingly prominent in time-series
 163 forecasting [? ?]. DeepAR provides a probabilistic recurrent alternative
 164 that learns across multiple related time series and produces predictive dis-
 165 tributions rather than only point forecasts [?]. Recent retail studies have
 166 also explored Temporal Fusion Transformers and other attention-based ar-
 167 chitectures on M5 and related datasets [? ? ?]. Recent C&IE work also
 168 shows the value of explicitly combining multiple time scales before forecast-
 169 ing, reinforcing the broader importance of temporal feature representation

170 in supply-chain demand prediction [?]. These approaches can be power-
 171 ful when data volume, sequence structure, and tuning resources justify their
 172 complexity [? ? ? ?]. However, their additional computational and mod-
 173 elling requirements can make them less attractive for smaller post-filtering
 174 samples or applications where transparent tabular feature effects are impor-
 175 tant. The present study consequently does not claim that tree-based models
 176 are universally superior to deep learning. Instead, it evaluates a deliberately
 177 bounded model set suited to the available M5 analytical sample and the
 178 study’s emphasis on reproducibility and decision translation.

179 *2.3. Feature Engineering and Multi-Source Data Integration*

180 Feature engineering transforms raw transactional and temporal informa-
 181 tion into variables that expose useful predictive structure to a learning al-
 182 gorithm. Lagged demand and rolling-window statistics are particularly im-
 183 portant because they encode recent demand history while avoiding the need
 184 for a model to infer every temporal relationship from raw dates. Reviews of
 185 supply-chain forecasting literature identify feature engineering and the inte-
 186 gration of temporal, pricing, and promotional information as recurring deter-
 187 minants of forecasting quality [? ? ?]. Systematic mapping work also notes
 188 that incomplete reporting of feature construction can make otherwise strong
 189 forecasting studies difficult to reproduce [?]. Feature attribution methods
 190 such as SHAP provide more formal explanations for tree-based predictions,
 191 while simpler impurity-based feature importance can provide a descriptive
 192 diagnostic [? ?]. Recent work also compares forecasting approaches un-
 193 der distributed learning constraints using Walmart sales data, illustrating
 194 that accuracy and computational efficiency can be considered jointly [?].
 195 The present study uses Random Forest impurity-based importance because
 196 the objective is to identify the relative contribution of engineered predic-
 197 tors rather than to provide a full local explanation framework. The results
 198 are interpreted cautiously because correlated lag and rolling features can
 199 share predictive information. This feature-centric design also provides a di-
 200 rect test of whether adding price and calendar variables improves a narrower
 201 historical-demand baseline.

202 *2.4. Forecast-Driven Inventory Optimisation*

203 The translation of forecasts into inventory decisions is rooted in classi-
 204 cal operations research. Safety stock provides a buffer against demand and
 205 forecast uncertainty, while service level represents the desired probability or

206 proportion of demand satisfied without stockout [? ?]. The newsvendor
 207 problem formalises the trade-off between ordering too little and ordering too
 208 much under uncertain demand [?]. Machine-learning research has increas-
 209 ingly revisited this problem by using predictive variables to estimate demand
 210 or order quantities directly [? ?]. Recent work has also embedded inventory
 211 cost objectives more directly into forecasting models, reinforcing the value of
 212 evaluating forecast accuracy alongside inventory consequences [? ?]. Recent
 213 empirical work using M5 has shown that forecast accuracy and inventory
 214 performance are related but not necessarily one-to-one [?]. Probabilistic
 215 forecasting research similarly argues that uncertainty information can be im-
 216 portant for downstream inventory planning [? ? ? ?]. The present study
 217 retains a transparent two-step forecast-then-translate approach rather than
 218 directly optimising the inventory decision. This design keeps the statistical
 219 accuracy measures and scenario calculations separately interpretable while
 220 still allowing their relationship to be examined.

221 *2.5. Hierarchical Structure and Robustness*

222 Retail demand data are naturally hierarchical, with products aggregated
 223 through categories, stores, and geographical levels. Hierarchical forecasting
 224 research has developed reconciliation methods that seek coherent forecasts
 225 across aggregation levels [? ?]. The present study does not perform hier-
 226 archical reconciliation and therefore does not claim coherence across the full
 227 M5 hierarchy. Instead, it focuses on 500 high-volume item–store series and
 228 examines robustness through demand-regime analysis and external valida-
 229 tion. This narrower design is appropriate for the research questions because
 230 it isolates the relationship between engineered features, forecast accuracy,
 231 and downstream decision scenarios. The limitation is that the results cannot
 232 automatically be transferred to slow-moving or intermittent-demand series.
 233 Research on intermittent demand and seasonal retail forecasting reinforces
 234 the importance of considering demand regime and data sparsity when eval-
 235 uating forecasting systems [? ?]. The robustness analyses in this study are
 236 therefore presented as empirical diagnostics rather than as proof of universal
 237 generalisation [? ?]. Recent lag-aware transformer research further illus-
 238 trates the continuing expansion of demand-forecasting methods beyond the
 239 bounded tabular model set used here [?].

Table 1: Comparative positioning of the present study relative to representative demand-forecasting and inventory-analytics research. The comparison is conceptual because the cited studies use different datasets, targets, horizons, and evaluation protocols.

Study	Focus	Data / Method	Main contribution	Difference from present study
OptimizedHyperparam[?]]	Retail forecasting	Tree-based methods with explanatory variables	Shows that simplified tree methods can be highly competitive	Does not provide the present study’s financial and safety-stock translation
Nsir2025XGBoost[?]]	Supply-chain prediction	Optimised XGBoost	Demonstrates high predictive accuracy after optimisation	Focuses on prediction rather than an integrated inventory-cost analysis
Bergsma2025InventoryLiterature review[?]]	Literature review	ML/DL supply-chain forecasting	Synthesises the role of multi-source features	Review-level contribution without a new empirical pipeline
Theodorou2025InventoryForecast-inventory relationship[?]]	Forecast-inventory relationship	M5 competition data	Shows that forecast accuracy and inventory performance are related non-linearly	Does not provide the same head-to-head three-model and uncertainty workflow
Long2025ProbRetail[?]]	Probabilistic retail forecasting	Gradient-boosted trees	Connects uncertainty-aware forecasting with retail applications	Uses a probabilistic framework rather than the present post-hoc error-based safety-stock analysis
Liu2025Stockout[?]]	Stockout prediction	Machine learning	Identifies near-term forecasting information as relevant to stockout prediction	Focuses on stockout prediction rather than revenue, holding cost, and safety stock together
Present study	Forecasting + decision translation	M5 top-500 series; LR, RF, XGBoost; supporting LightGBM/ARIMA	Unifies accuracy, financial, inventory, uncertainty, regime, and external-validation analyses	Single benchmark and bounded modelling sample; scenario costs and lead time are illustrative

3. Methodology

3.1. Detailed Methodology

The research design combines data selection, multi-source integration, feature engineering, model training, chronological validation, and downstream financial and inventory analysis within a single executable notebook. The notebook was executed in a Kaggle environment using Python and standard open-source data-science libraries. The three core models are kept separate from the supporting LightGBM and ARIMA analyses. All headline M5 models use the same training observations, feature matrix, and 28-day test horizon. The downstream decision analyses use the same M5 test observations and LightGBM predictions rather than introducing a second test sample. This design ensures that the accuracy and decision-oriented layers remain connected while their purposes remain distinct. The workflow is shown in Figure 1. The external DataCo experiment is treated as a robustness check rather than as a directly comparable benchmark [?].

3.2. Dataset

The primary empirical source is the M5 Forecasting Accuracy competition dataset [?]. The dataset contains daily Walmart unit sales for 3,049 products

Table 2: Raw M5 data files used before preprocessing and model-series selection.

File	Rows	Columns
Sales ('sales_train_validation.csv')	30,490	1,919
Calendar ('calendar.csv')	1,969	14
Prices ('sell_prices.csv')	6,841,121	4

across ten stores in California, Texas, and Wisconsin, represented as 30,490 item-store time series. The raw sales table contains 30,490 rows and 1,919 columns, while the calendar and selling-price tables contain 1,969 rows by 14 columns and 6,841,121 rows by 4 columns, respectively. The analysis selects the 500 highest-volume item-store series using only training-period sales. This prevents test-period demand from influencing series selection. The selected series are reshaped into 956,500 product-day observations before calendar and price information are merged. The final model-ready feature matrix contains ten numerical predictors.

An independent DataCo SMART SUPPLY CHAIN dataset is used for external validation, where transactions are aggregated to daily order counts and a separate chronological 80/20 evaluation is performed [?].

3.3. Data Preprocessing

Date fields are converted to datetime format and decomposed into year, month, ISO week, day of week, and quarter. For each item-store series, 7-day and 28-day lag variables and 7-day and 28-day trailing means are constructed. The rolling means are shifted by one day before the rolling operation so that the current target is not included in its own predictors. Selling prices are forward-filled and backward-filled within each item-store series before the period-over-period price-change feature is calculated. Rows are not removed merely because event-name or event-type information is unavailable in the final model matrix. Instead, only rows lacking required numerical modelling variables are removed. The resulting structural loss occurs at the beginning of each series because lag and rolling variables require historical observations. The initial 956,500 product-day observations are reduced to 942,500 analytical rows.

3.4. Feature Engineering

The final feature matrix contains ten numerical predictors grouped into three information families. Historical-demand features are `lag_7`, `lag_28`,

287 `rolling_mean_7`, and `rolling_mean_28`. Pricing features are `sell_price`
288 and `price_change`. Calendar-derived features are `month`, `week`, `dayofweek`,
289 and `quarter`. Product and store identifiers are retained for grouping and
290 aggregation but are not supplied directly as predictors. RQ1 evaluates pro-
291 gressively richer feature configurations. RQ2 uses Random Forest impurity-
292 based feature importance to identify the relative contribution of the engi-
293 neered variables. Because several lag and rolling features are correlated,
294 importance values are interpreted as descriptive rather than causal measures
295 of information contribution.

296 *3.5. Forecasting Models*

297 Linear Regression provides the simplest baseline and models demand as
298 a weighted combination of the engineered predictors. Random Forest uses
299 50 trees, maximum depth 16, minimum leaf size 2, a 50% bootstrap fraction,
300 random seed 42, and two parallel jobs. XGBoost uses 100 estimators, learning
301 rate 0.1, maximum depth 6, random seed 42, and a squared-error regression
302 objective. LightGBM is used as a supporting model with 100 estimators and
303 random seed 42 for M5 downstream analyses. ARIMA(5,1,0) is used only as
304 a supporting statistical benchmark on the aggregate daily series. The three
305 core models are trained on the same ten-feature matrix and evaluated on
306 exactly the same test observations. LightGBM and ARIMA are therefore not
307 mixed into the headline three-model ranking. This separation is maintained
308 throughout the Results and Discussion sections.

309 *3.6. Experimental Settings*

310 The experimental settings were fixed before the final chronological evalu-
311 ation so that the three core models were compared under identical data and
312 test conditions. Random Forest, XGBoost, and LightGBM used the config-
313 urations reported in Table 3. The supporting ARIMA analysis was fitted
314 only to the aggregate daily series and is not included in the row-level model
315 ranking.

316 *3.7. Chronological Validation*

317 The final analytical dataset is ordered by date and divided chronologically.
318 The last 28 calendar days, from 28 March 2016 through 24 April 2016, form
319 the test horizon. The preceding observations form the training period from
320 26 February 2011 through 27 March 2016. This produces 928,500 training
321 rows and 14,000 test rows, corresponding to 500 selected item–store series

Table 3: Model configurations used in the experiments. LightGBM and ARIMA are supporting models and are not part of the headline three-model comparison.

Parameter	Random Forest	XGBoost	LightGBM
Number of estimators	50	100	100
Learning rate	–	0.1	Default
Maximum tree depth	16	6	Default
Minimum samples per leaf	2	–	–
Bootstrap fraction	0.5	–	–
Random seed	42	42	42

over 28 test days. A chronological split is essential because a random row-wise split could allow information from later dates to influence the evaluation of earlier dates. The 28-day horizon also mirrors the forecasting horizon of the M5 competition. All three core models are therefore evaluated under the same temporal conditions.

3.8. Evaluation Metrics

Forecast accuracy is evaluated using MAE, RMSE, and the coefficient of determination R^2 . For observed demand y_i and predicted demand $\hat{y}_i$ over n test observations, MAE is defined as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (1)$$

RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (2)$$

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (3)$$

MAE measures average absolute forecast error in demand units, RMSE gives greater weight to large errors, and R^2 describes the proportion of observed demand variance explained by the model [?]. The three measures are

336 reported together because they capture different characteristics of the error
 337 distribution.

338 *3.9. Financial Translation*

339 The downstream financial analysis uses LightGBM predictions from the
 340 same 14,000 M5 test observations. Fulfilled-demand revenue is calculated
 341 as the minimum of actual and forecast demand multiplied by selling price.
 342 Stockout-cost exposure is calculated as unmet demand multiplied by selling
 343 price and a 10% cost factor. Holding-cost exposure is calculated as over-
 344 forecast demand multiplied by selling price and a 5% cost factor. These
 345 percentages are illustrative assumptions rather than observed Walmart ac-
 346 counting parameters. The resulting dollar quantities are therefore scenario
 347 outputs rather than estimates of actual Walmart profit, revenue, or operating
 348 cost. This distinction is essential because the public M5 dataset does not con-
 349 tain retailer-specific procurement, shortage, holding, labour, or logistics cost
 350 structures. The financial analysis is intended to demonstrate how forecast
 351 errors can be translated into decision-oriented quantities under transparent
 352 assumptions. It does not constitute a complete profit optimisation model.

353 *3.10. Inventory and Uncertainty Analysis*

354 Safety stock is estimated using

$$SS = z_{\alpha}\sigma_e\sqrt{L}, \quad (4)$$

355 where z_{α} is the standard-normal quantile associated with the target ser-
 356 vice level, σ_e is the standard deviation of signed forecast error, and L is the
 357 assumed lead time. The main service-level analysis evaluates targets from
 358 70% to 95% using a seven-day lead time. The uncertainty analysis divides
 359 the 14,000 observations into three equal-sized tertiles using absolute fore-
 360 cast error. Tier-specific safety stock is then calculated using each tier’s error
 361 standard deviation at a 95% service target and seven-day lead time. This
 362 procedure links inventory buffers directly to observed forecast uncertainty. It
 363 remains a simplified scenario analysis because actual lead-time distributions,
 364 replenishment constraints, service-level definitions, and product substitution
 365 are not available in M5.

366 3.11. External Validation

367 External validation uses the DataCo SMART SUPPLY CHAIN FOR BIG
368 DATA ANALYSIS dataset. The structured data contain 180,519 transaction
369 rows in the notebook. Transactions are aggregated to daily order counts
370 and four historical-demand features are constructed. LightGBM is evaluated
371 using a chronological 80/20 split. The experiment produces MAE 50.32 and
372 RMSE 61.25. Because the target scale and demand-generating process differ
373 from M5, these metrics are not compared numerically with the M5 metrics.
374 The purpose of the experiment is instead to determine whether the basic lag
375 and rolling feature-engineering strategy remains usable outside the original
376 benchmark. The external validation is therefore treated as a robustness check
377 rather than evidence of universal generalisation.

378 3.12. Research Ethics and Reproducibility

379 The study uses publicly available benchmark and research datasets and
380 does not involve human participants, personally identifiable information, or
381 experimental intervention. Reproducibility is supported by the single note-
382 book workflow, explicit feature definitions, fixed random seeds for stochastic
383 models, and the fixed chronological hold-out. The software environment uses
384 Python with pandas, NumPy, scikit-learn, XGBoost, LightGBM, statsmod-
385 els, matplotlib, and seaborn. The model configurations and evaluation design
386 are reported explicitly. The source notebook and repository should accom-
387 pany the manuscript according to the journal and institutional submission
388 requirements.

389 4. Results

390 All headline M5 results use the same chronological 28-day test horizon
391 containing 14,000 item-store-day observations. The core comparison con-
392 sists of Linear Regression, Random Forest, and XGBoost. LightGBM and
393 ARIMA are reported only as supporting diagnostics. The downstream fi-
394 nancial, inventory, uncertainty, and demand-regime analyses use LightGBM
395 predictions from the same M5 test observations.

396 4.1. RQ1: Overall Forecasting Accuracy

397 RQ1 evaluates how forecasting accuracy changes as historical, pricing,
398 and calendar information are combined. Table 4 and Figure 4 report the

Table 4: Forecast MAE by feature configuration on the 14,000-observation chronological M5 test set. Lower values indicate lower absolute forecast error.

Feature configuration	MAE
Baseline: lag and rolling-window features	5.842
Price-enhanced	5.836
Calendar-enhanced	6.691
Full multi-source	5.608

four feature configurations and their corresponding MAE values. Four feature configurations are compared using Random Forest: a lag-and-rolling baseline, a price-enhanced configuration, a calendar-enhanced configuration, and the full multi-source feature matrix. The baseline obtains MAE 5.842. Adding price information alone changes MAE only slightly to 5.836. The calendar-enhanced configuration performs worse, with MAE 6.691. The full multi-source configuration achieves the lowest MAE at 5.608, representing approximately a 4.0% improvement over the baseline. These results show that the combined feature matrix performs best in the present experiment, while the magnitude of the improvement is primarily associated with the historical demand features.

4.2. RQ2: Contribution of Engineered Features

RQ2 examines which engineered variables contribute most strongly to the Random Forest predictions. Table 5 and Figure 5 report the numerical importance scores and their ranking. The 7-day rolling mean has an importance score of 0.7991, equivalent to approximately 79.9% of the reported importance. The 28-day rolling mean has importance 0.0516. Day of week, lag 7, lag 28, week, selling price, month, and price change have importance values of 0.0408, 0.0306, 0.0289, 0.0208, 0.0202, 0.0069, and 0.0010, respectively. The two rolling-window variables together account for approximately 85.5% of the importance diagnostic. The correlation analysis shows strong relationships among lag and rolling-window predictors. This means that the importance values should not be interpreted as independent causal contributions. Nevertheless, the ranking indicates that recent demand history provides the dominant predictive signal in the selected feature matrix.

Table 5: Random Forest impurity-based feature-importance scores for the variables included in the RQ2 diagnostic. The values are descriptive importance measures and should not be interpreted as causal effects.

Feature	Importance
rolling_mean_7	0.7991
rolling_mean_28	0.0516
dayofweek	0.0408
lag_7	0.0306
lag_28	0.0289
week	0.0208
sell_price	0.0202
month	0.0069
price_change	0.0010

4.3. RQ3: Comparative Model Evaluation

RQ3 compares Linear Regression, Random Forest, and XGBoost under identical training and test conditions. Table 6 and Figure 6 report the core-model results on the identical test observations. XGBoost achieves the strongest core performance, with MAE 5.582, RMSE 8.690, and $R^2 = 0.705$. Random Forest follows closely with MAE 5.608, RMSE 8.753, and $R^2 = 0.701$. Linear Regression records MAE 5.935, RMSE 9.141, and $R^2 = 0.674$. The supporting LightGBM model achieves MAE 5.584 and RMSE 8.693 on the same row-level M5 observations. ARIMA achieves MAE 958.511 and RMSE 1,107.919 after the test observations are aggregated to the daily series, so these values are not directly comparable with the item-store row-level metrics. The core ranking therefore identifies XGBoost as the strongest model among the three primary candidates for this feature matrix and horizon. The small difference between XGBoost and Random Forest also indicates that the result should not be interpreted as evidence that greater model complexity is universally superior.

4.4. RQ4: Financial Implications of Forecasting

RQ4 translates the LightGBM forecasts into scenario-level financial quantities using the same 14,000 M5 test observations. Table 7 and Figure 7 report the resulting revenue, stockout-cost, and holding-cost quantities. Fulfilled-demand revenue is approximately \$335,984.96. Stockout-cost exposure is

Table 6: Core model performance on the identical 14,000-observation chronological M5 test set. The supporting LightGBM result is reported separately because it is not part of the headline three-model comparison.

Model	MAE	RMSE	R^2
Linear Regression	5.935	9.141	0.674
Random Forest	5.608	8.753	0.701
XGBoost	5.582	8.690	0.705
Supporting LightGBM	5.584	8.693	–

Table 7: Financial implications calculated from LightGBM forecasts over the 14,000-observation M5 test horizon under illustrative unit-cost assumptions.

Metric	Value (USD)
Fulfilled-demand revenue	335,984.96
Stockout-cost exposure	6,020.06
Holding-cost exposure	3,273.91

approximately \$6,020.06. Holding-cost exposure is approximately \$3,273.91. The stockout-cost total is approximately 1.84 times the holding-cost total under the specified assumptions. This ratio is not a general empirical retail finding because it depends on the realised errors, selling prices, and the assumed 10% and 5% cost factors. The analysis therefore demonstrates the mechanics of financial translation rather than estimating retailer-specific profitability. Procurement, labour, logistics, and other business costs are outside the calculation. The quantities should consequently be interpreted as transparent scenarios generated from held-out forecasts.

4.5. RQ5: Inventory Service-Level Trade-off

RQ5 evaluates how estimated safety stock changes with the target service level. Table 8 and Figure 8 report the six service-level scenarios from 70% to 95%. Using the observed LightGBM forecast-error dispersion and a seven-day lead time, estimated safety stock increases from 12.0 units at 70% service to 37.8 units at 95% service. The intermediate estimates are 15.5, 19.3, 23.8, and 29.4 units at service levels of 75%, 80%, 85%, and 90%, respectively. The relationship is monotonic because higher target service levels require larger buffers against forecast uncertainty. The increase is not linear, reflecting

Table 8: Estimated safety-stock requirement by target service level using the observed M5 LightGBM forecast-error dispersion and a seven-day lead-time assumption.

Target service level (%)	Safety stock (units)
70	12.0
75	15.5
80	19.3
85	23.8
90	29.4
95	37.8

the behaviour of the normal quantile in the safety-stock formulation. The associated reorder-point simulation produces an illustrative reorder point of approximately 148.4 units under the 95% service scenario. Because the seven-day lead time is assumed rather than observed from a retailer replenishment system, these quantities are scenario estimates. They should therefore not be interpreted as production inventory recommendations.

4.6. RQ6: Forecast Uncertainty and Stockout Risk

RQ6 divides the 14,000 test observations into three equal-sized uncertainty tertiles using absolute forecast error. Table 9 and Figure 9 report the tier-specific error and safety-stock results. The low-, medium-, and high-uncertainty groups contain 4,667, 4,666, and 4,667 observations, respectively. Mean absolute errors are 1.036, 3.850, and 11.866 units across the three groups. Using the corresponding error standard deviations, a 95% service target, and a seven-day lead time produces safety-stock estimates of 2.74, 4.36, and 36.05 units. The high-uncertainty group therefore requires a substantially larger illustrative buffer than the other groups. The operational risk zones contain 8,541 low-risk, 4,534 medium-risk, and 925 high-risk observations under the notebook thresholds. The relationship between absolute forecast error and stockout risk is mechanically positive because stockout risk is defined using the under-forecast component of demand error. The result is therefore interpreted as a risk diagnostic rather than evidence of a causal relationship.

4.7. RQ7: Robustness Across Demand Regimes

RQ7 evaluates whether forecast performance remains stable across low-, medium-, and high-demand regimes. Table 10 and Figure 10 report the

Table 9: Forecast uncertainty and error-based safety-stock estimates by uncertainty tertile. Safety stock uses a 95% service target and seven-day lead time.

Tier	n	Mean abs. error	Std. dev.	Safety stock
Low	4,667	1.036	0.631	2.74
Medium	4,666	3.850	1.001	4.36
High	4,667	11.866	8.283	36.05

Table 10: LightGBM forecast accuracy by demand regime on the M5 test observations and the separate DataCo external-validation result. The DataCo metrics are reported on their own target scale and are not numerically compared with M5.

Analysis	Group / Dataset	MAE	RMSE
Demand regime	Low demand	3.595	—
Demand regime	Medium demand	5.155	—
Demand regime	High demand	8.023	—
External validation	DataCo SMART SUPPLY CHAIN	50.32	61.25

regime-level MAE results, while the table also records the separate DataCo robustness check. The 500 selected series are divided using the training-period 28-day rolling-demand baseline. LightGBM MAE increases from 3.595 in the low-demand regime to 5.155 in the medium-demand regime and 8.023 in the high-demand regime. The high-demand regime therefore has more than twice the MAE of the low-demand regime. The result shows that residual forecasting error is not distributed uniformly across the selected portfolio. Higher-volume item-store combinations deserve particular attention when uncertainty-aware inventory policies are designed. The rolling-MAE and two-dimensional density diagnostics support the same pattern without being interpreted as evidence of long-term deployment stability. The conceptual scalability analysis indicates that the modular pipeline can be repeated for more forecasting entities, but no measured production-scale runtime benchmark is claimed. The separate DataCo external validation produces MAE 50.32 and RMSE 61.25 on its own target scale; these values are reported in Table 10 as a robustness check rather than as a direct comparison with M5.

4.8. Comparison with Existing Techniques

The reported contemporary studies use different datasets, targets, forecast horizons, and evaluation protocols, so their numerical accuracy values

Table 11: Reported accuracy of selected contemporary demand-forecasting studies alongside the present study. Values are contextual because datasets, targets, horizons, and evaluation protocols differ.

Study	Dataset scope	Best method	Reported result
Deepak2025RFLines [?]	Supply-chain de- mand	Uni-regression	MAE 1.73, RMSE 2.03, $R^2 = 0.94$
Nsir2025XGBoost [?]	Industry 4.0 supply chain	BiLSTM/NARX Optimised XG- Boost	$R^2 = 0.985$
Long2025ProbRetail [?]	M5, Favorita, e- commerce	Probabilistic gradient-boosted trees	Dataset-dependent scaled pin- ball loss
Theodorou2025Inven- tory [?]	M5 competition	Multiple methods	Forecast-inventory relation- ship
Present study	M5 top-500 item- store series	XGBoost	MAE 5.582, RMSE 8.690, $R^2 = 0.705$

are contextual rather than head-to-head comparisons [? ? ? ?]. A supply-chain BiLSTM/NARX study reports MAE 1.73, RMSE 2.03, and $R^2 = 0.94$ on its own dataset [?]. An optimised XGBoost study reports $R^2 = 0.985$ for sustainable supply-chain performance prediction [?]. Probabilistic gradient-boosted approaches have been evaluated across M5, Favorita, and proprietary retail data using scaled probabilistic losses rather than the point-error metrics used here [?]. M5-based work has also demonstrated that forecast accuracy and inventory performance can be related without being identical [?]. The present study does not claim state-of-the-art forecasting accuracy from these comparisons [?]. Its contribution is instead the explicit connection of a reproducible forecasting pipeline to financial, inventory, uncertainty, demand-regime, and external-validation analyses using the same primary test observations. This positioning is consistent with the study’s objective of decision-oriented transparency rather than leaderboard optimisation.

5. Discussion

5.1. RQ1: Feature Configuration

The RQ1 results show that combining historical, pricing, and calendar information produced the lowest MAE among the four tested configurations. The improvement from 5.842 to 5.608 is modest in absolute terms, while the calendar-only configuration deteriorates to 6.691. This indicates that additional variables do not automatically improve a forecasting pipeline. The

529 result is more consistent with a feature-quality interpretation in which re-
 530 cent demand history provides the strongest signal and other sources add
 531 only limited incremental information. The finding is relevant to structured
 532 retail forecasting because it supports a transparent feature-engineering ap-
 533 proach rather than a simple increase in feature count. It also provides a clear
 534 empirical basis for retaining the full feature configuration for the subsequent
 535 model comparison. The conclusion is limited to the selected high-volume M5
 536 series and the fixed chronological horizon used in this study.

537 5.2. RQ2: Engineered Feature Contribution

538 The RQ2 importance diagnostic shows that the 7-day rolling mean ac-
 539 counts for 79.9% of the reported Random Forest importance, with the two
 540 rolling means together accounting for approximately 85.5%. This concen-
 541 tration indicates that recent demand history dominates the predictive signal
 542 in the selected feature matrix. Price and calendar variables remain present
 543 but contribute much smaller shares. The result should not be interpreted as
 544 causal evidence because lagged and rolling predictors are correlated. Instead,
 545 the importance ranking is useful as a model diagnostic that identifies which
 546 information the fitted Random Forest relies on most strongly. This provides
 547 a practical basis for monitoring the stability of temporal features when the
 548 pipeline is extended to other retail settings. The finding also supports the
 549 decision to keep the model architecture relatively transparent rather than
 550 relying on a larger black-box feature space.

551 5.3. RQ3: Core Model Comparison

552 XGBoost provides the strongest row-level performance among the three
 553 core models, but its advantage over Random Forest is narrow. The MAE
 554 difference is only 0.026 demand units, while both ensemble methods out-
 555 perform Linear Regression on MAE and RMSE. LightGBM produces a very
 556 similar result and was therefore suitable for the downstream analyses. The
 557 supporting ARIMA result is not directly comparable because it is fitted after
 558 aggregation to a daily series. The evidence therefore supports XGBoost as
 559 the strongest candidate for the specific feature matrix and horizon rather
 560 than as universally superior to other forecasting families. This distinction is
 561 important because model rankings can change with demand structure, ag-
 562 gregation level, horizon, and feature availability. The result is consequently
 563 best viewed as an empirical ranking under controlled conditions.

564 5.4. RQ4: Financial Translation

565 The RQ4 results demonstrate how the same held-out forecasts can be
566 translated into scenario-level financial quantities. Fulfilled-demand revenue
567 is approximately \$335,984.96, while stockout-cost and holding-cost exposure
568 are approximately \$6,020.06 and \$3,273.91. The stockout-to-holding ratio is
569 driven by the realised errors and the illustrative 10% and 5% cost factors.
570 It therefore cannot be treated as a general retail cost relationship. The
571 contribution of this analysis is the explicit connection between forecast output
572 and decision-oriented financial quantities using the same test observations.
573 This makes the operational meaning of the forecast easier to inspect than a
574 model score alone. At the same time, retailer-specific costs would be required
575 before these quantities could support a production financial decision.

576 5.5. RQ5: Service-Level Trade-off

577 The RQ5 results show a clear increase in estimated safety stock as the tar-
578 get service level rises. The estimated requirement increases from 12.0 units
579 at 70% service to 37.8 units at 95% service, with intermediate values in-
580 creasing monotonically. The non-linear increase reflects the normal-quantile
581 component of the safety-stock formulation. The result makes the service-
582 level and inventory-investment trade-off explicit within the same forecasting
583 framework. The seven-day lead time is assumed, so the values should be
584 interpreted as scenario estimates rather than observed retailer policy param-
585 eters. Nevertheless, the pattern shows why a forecast can be operationally
586 useful even when its accuracy is not perfect, provided its uncertainty is trans-
587 lated into a transparent inventory assumption. The analysis therefore com-
588 plements the predictive comparison with a decision-oriented view of forecast
589 value.

590 5.6. RQ6: Forecast Uncertainty and Stockout Risk

591 The RQ6 results show that forecast uncertainty is highly uneven across
592 the M5 test observations. The high-uncertainty tertile has MAE 11.866 and
593 estimated safety stock 36.05 units, compared with 1.036 and 2.74 units in
594 the low-uncertainty tertile. This supports differentiated inventory buffers
595 rather than a uniform allowance across all observations. The stockout-risk
596 relationship is mechanically positive because stockout risk is defined from the
597 under-forecast component of demand error. It is therefore a diagnostic rela-
598 tionship rather than evidence of a causal effect. The practical implication is

that forecast-risk diagnostics can help identify observations for which inventory protection should be reviewed more closely. The result also reinforces the importance of separating empirical risk diagnostics from prescriptive inventory policy.

5.7. *RQ7: Demand Regimes and External Robustness*

The RQ7 results show that MAE increases from 3.595 in the low-demand regime to 8.023 in the high-demand regime. This indicates that forecast error is not uniform across the selected portfolio and that higher-demand series deserve particular attention when uncertainty-aware policies are considered. The separate DataCo validation produces MAE 50.32 and RMSE 61.25 on a different target scale. That result supports cautious transferability of the temporal feature-engineering strategy but does not establish cross-dataset metric equivalence or universal generalisation. Taken together, the regime and external-validation results indicate that robustness should be examined both within the primary portfolio and across a structurally different dataset. The conceptual scalability analysis further suggests that the modular pipeline can be repeated for more forecasting entities, although production-scale runtime has not been measured. This keeps the interpretation aligned with the evidence actually generated by the notebook.

5.8. *Managerial Implications*

For retail planners, the findings support keeping recent demand history central to short-horizon forecasting while validating the incremental value of price and calendar variables. The core model comparison indicates that a standard tree-based pipeline can provide strong predictive performance without requiring a deep-learning architecture. The financial and safety-stock analyses show that forecasts should be accompanied by explicit assumptions about cost, lead time, and target service level before being converted into inventory decisions. The uncertainty and demand-regime diagnostics can help prioritise observations where forecast risk is comparatively high. The framework can therefore serve as a transparent decision-support layer rather than a fully prescriptive optimiser. A production implementation would still require retailer-specific costs, lead-time distributions, replenishment constraints, substitution effects, and rolling-origin validation. This interpretation is consistent with the study’s objective of connecting predictive evidence to operational decision support without overstating the scope of the benchmark.

635 5.9. Limitations

636 The first limitation is the deliberate restriction to the 500 highest-volume
637 item–store series. The findings should not be automatically generalised to the
638 full 30,490-series M5 hierarchy or to slow-moving and intermittent-demand
639 products. Second, the financial translation uses illustrative cost factors rather
640 than retailer-specific procurement, shortage, holding, or lost-sales costs. Con-
641 sequently, the reported dollar quantities are scenario values rather than ob-
642 served Walmart financial performance. Third, the safety-stock calculations
643 rely on an assumed seven-day lead time and simplified service-level formula-
644 tion. Actual replenishment lead-time distributions, constraints, substitution
645 effects, and service definitions are not observed in M5. Fourth, the Dat-
646 aCo experiment uses a different target scale and feature structure and is
647 therefore not a controlled cross-dataset benchmark. Finally, the scalability
648 analysis is conceptual because the study does not report measured training
649 time, memory consumption, or production deployment performance for very
650 large numbers of forecasting entities.

651 5.10. Future Directions

652 Future research should extend the modelling sample beyond the top-500
653 high-volume series and evaluate whether the feature-engineering strategy re-
654 mains effective for intermittent and low-volume demand. A second direction
655 is probabilistic forecasting, where prediction intervals or quantile forecasts
656 can replace post-hoc uncertainty analysis and feed directly into inventory
657 decisions. A third direction is a richer inventory model using empirical lead-
658 time distributions, retailer-specific cost parameters, service-level constraints,
659 and replenishment policies. Hierarchical forecast reconciliation could also
660 be incorporated to test coherence across product, category, store, and geo-
661 graphical levels [? ? ?]. Measured computational benchmarks and rolling-
662 origin evaluation across multiple forecast windows would provide stronger
663 evidence for production-scale deployment and temporal robustness. Multi-
664 dataset validation could distinguish general feature-engineering effects from
665 benchmark-specific behaviour. Direct predictive-to-prescriptive optimisation
666 is another potential extension because forecast-then-optimize pipelines can
667 leave decision value on the table [?]. These extensions would move the
668 framework from transparent academic scenario analysis toward more com-
669 plete operational decision support.

670 6. Conclusion

671 This study developed a reproducible machine-learning framework that
672 connects retail demand forecasting with financial and inventory scenario
673 analysis. Using the M5 Forecasting dataset, the analysis selected the 500
674 highest-volume item-store series based only on training-period demand and
675 evaluated a chronological 28-day hold-out containing 14,000 observations.
676 The full multi-source feature configuration achieved Random Forest MAE
677 5.608 compared with 5.842 for the lag-and-rolling baseline, while the 7-day
678 rolling mean dominated the feature-importance diagnostic. Among the three
679 core models, XGBoost achieved the strongest row-level performance with
680 MAE 5.582, RMSE 8.690, and $R^2 = 0.705$. LightGBM produced a similar
681 row-level result and was used for downstream decision-oriented analyses.
682 Under illustrative cost assumptions, the forecasts corresponded to approx-
683 imately \$335,984.96 in fulfilled-demand revenue, \$6,020.06 in stockout-cost
684 exposure, and \$3,273.91 in holding-cost exposure. Estimated safety stock
685 increased from 12.0 units at 70% service to 37.8 units at 95%, while uncer-
686 tainty and demand-regime analyses showed that forecast risk is not uniform
687 across the portfolio. External validation on DataCo produced MAE 50.32
688 and RMSE 61.25 on its own scale, supporting cautious transferability of the
689 feature-engineering approach. Overall, the main contribution is not a claim
690 of universal model superiority but a transparent framework that evaluates
691 forecast accuracy and its operational implications from the same experimen-
692 tal evidence.

693 Declarations

694 *CRedit authorship contribution statement*

695 Mohamed Meraj Mansur Sharif: Conceptualization, Methodology, Soft-
696 ware, Data curation, Formal analysis, Visualization, Writing – original draft.
697 Raja Hashim Ali: Supervision, Validation, Writing – review and editing.
698 Talha Ali Khan: Supervision, Validation, Writing – review and editing.

699 *Funding*

700 No external funding was reported for this study.

701 *Declaration of Competing Interest*

702 The authors declare that they have no known competing financial or per-
703 sonal relationships that could have appeared to influence the work reported
704 in this paper.

705 *Data Availability*

706 The study uses publicly available benchmark and research datasets. The
707 M5 Forecasting Accuracy dataset is publicly distributed for forecasting re-
708 search and is available through the M5 competition data repository (<https://www.kaggle.com/c/m5-forecasting-accuracy/data>). The DataCo SMART
709 SUPPLY CHAIN dataset used for external validation is publicly available
710 through Mendeley Data (DOI: 10.17632/8gx2fvg2k6.5; [https://doi.org/](https://doi.org/10.17632/8gx2fvg2k6.5)
711 10.17632/8gx2fvg2k6.5).
712 10.17632/8gx2fvg2k6.5).

713 *Code Availability*

714 The experimental workflow was implemented in a Jupyter notebook exe-
715 cuted in a Kaggle environment. The reproducibility notebook, figure-generation
716 sources, and supporting LaTeX materials are included in the submission
717 package accompanying this manuscript.

718 *Declaration of Generative AI and AI-Assisted Technologies*

719 Generative artificial-intelligence tools were used during preparation of
720 the manuscript for language editing, restructuring and formatting assistance,
721 drafting and revising selected passages, and assistance with code and note-
722 book debugging. The underlying empirical work, data processing, model
723 execution, figures, exported tables, and interpretation were checked against
724 the project notebook and source files. The authors remain responsible for the
725 final wording, accuracy of the reported results, validity of the cited sources,
726 and compliance with applicable academic-integrity and publication require-
727 ments.

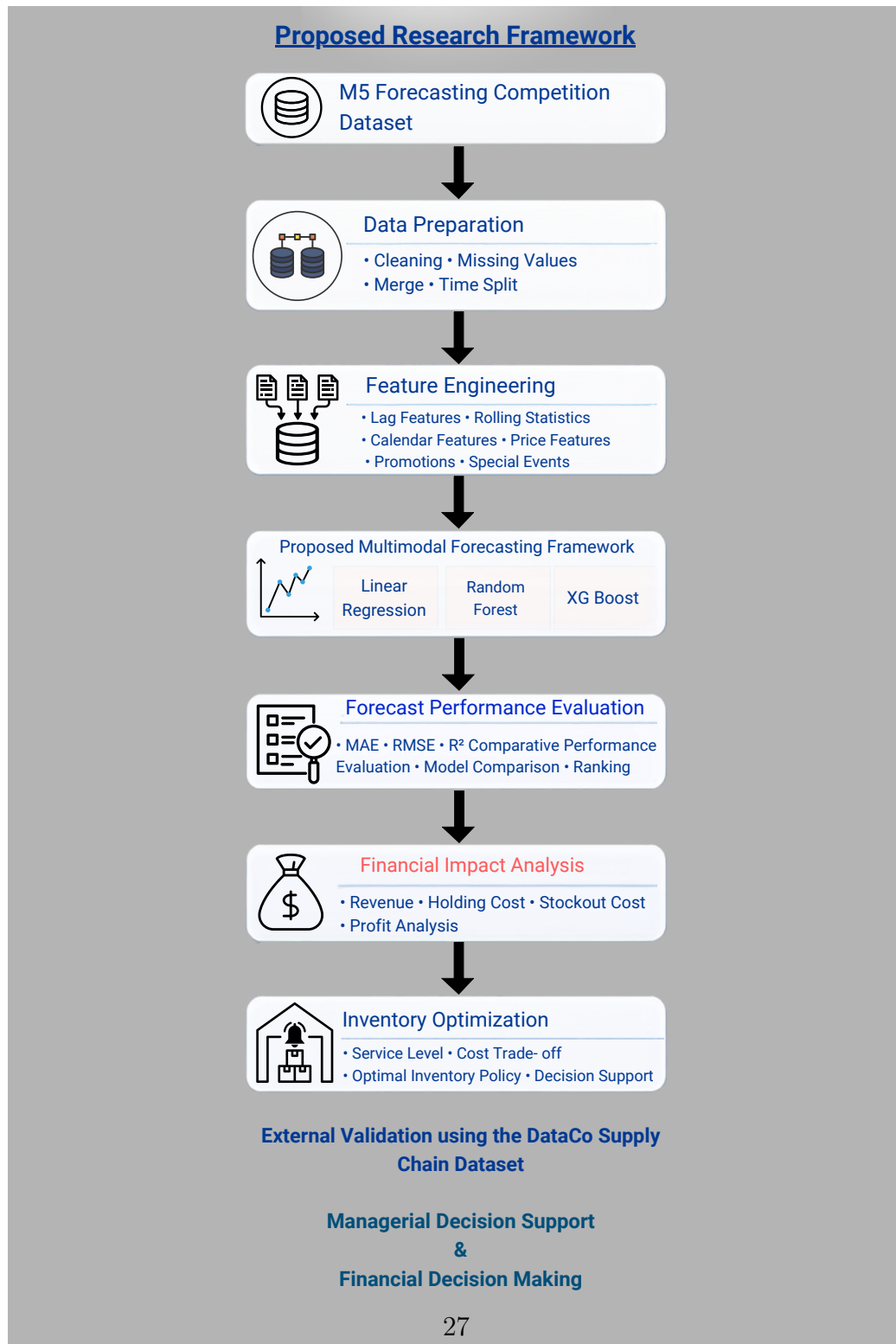


Figure 1: Overall research workflow used in the study, showing M5 series selection using training-period demand, data integration, preprocessing, lag and rolling feature engineering, chronological model training and validation, financial and inventory scenario analysis, uncertainty and demand-regime diagnostics, and external validation using the DataCo SMART SUPPLY CHAIN dataset.

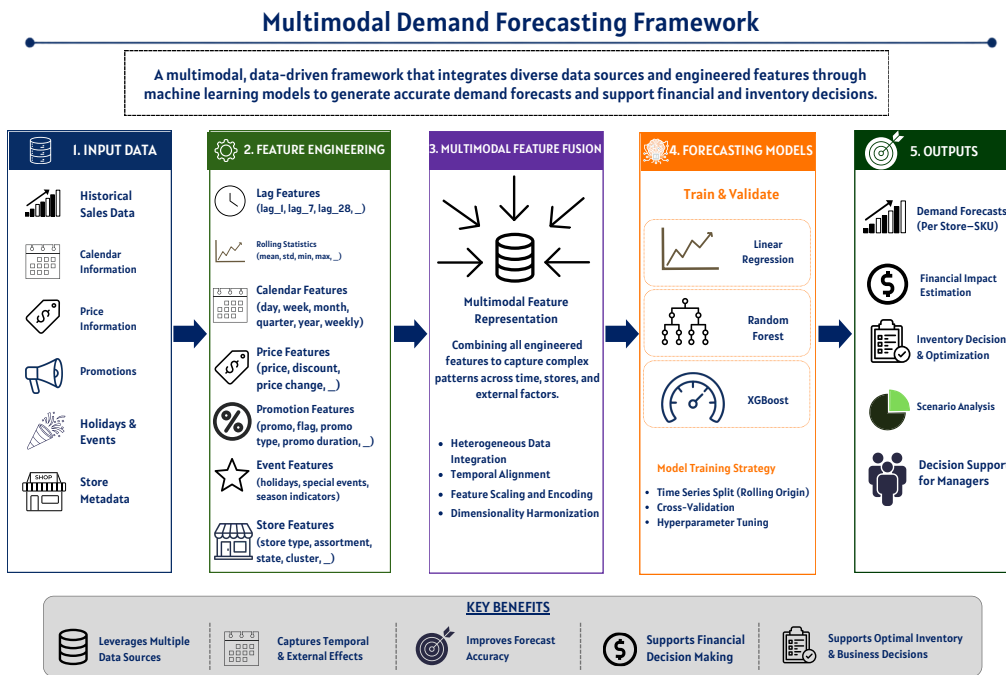


Figure 2: Multi-source forecasting framework linking heterogeneous retail inputs, feature engineering, model evaluation, and decision-oriented financial and inventory analysis.

Figure 3. M5 Forecasting Dataset Architecture

The M5 forecasting dataset integrates multiple heterogeneous data sources to provide a comprehensive representation of retail demand across products, stores, and time, enabling robust multimodal demand forecasting.

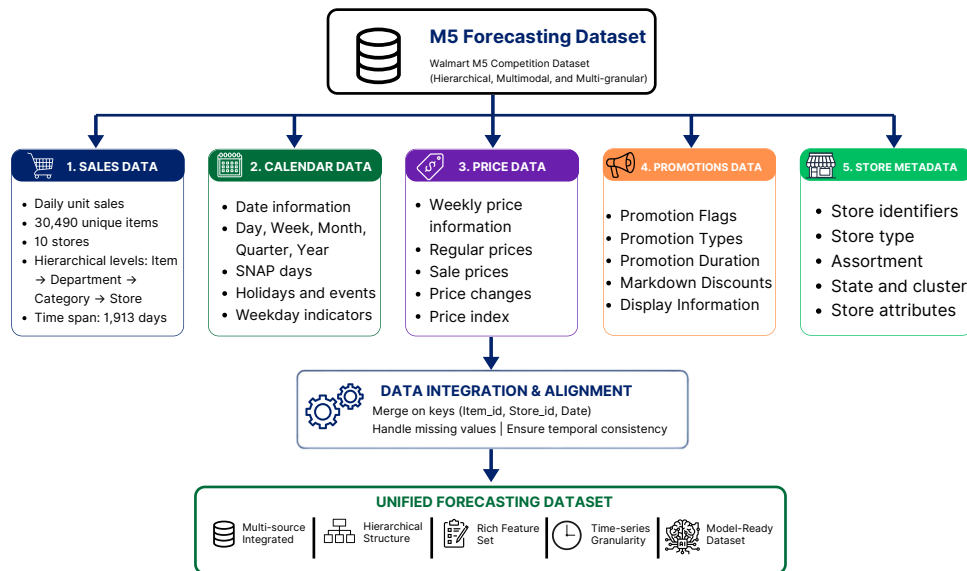


Figure 3: M5 dataset architecture and multi-source data integration used in the empirical pipeline. The diagram distinguishes the sales, calendar, and price sources from the unified analytical representation used for feature engineering and forecasting.

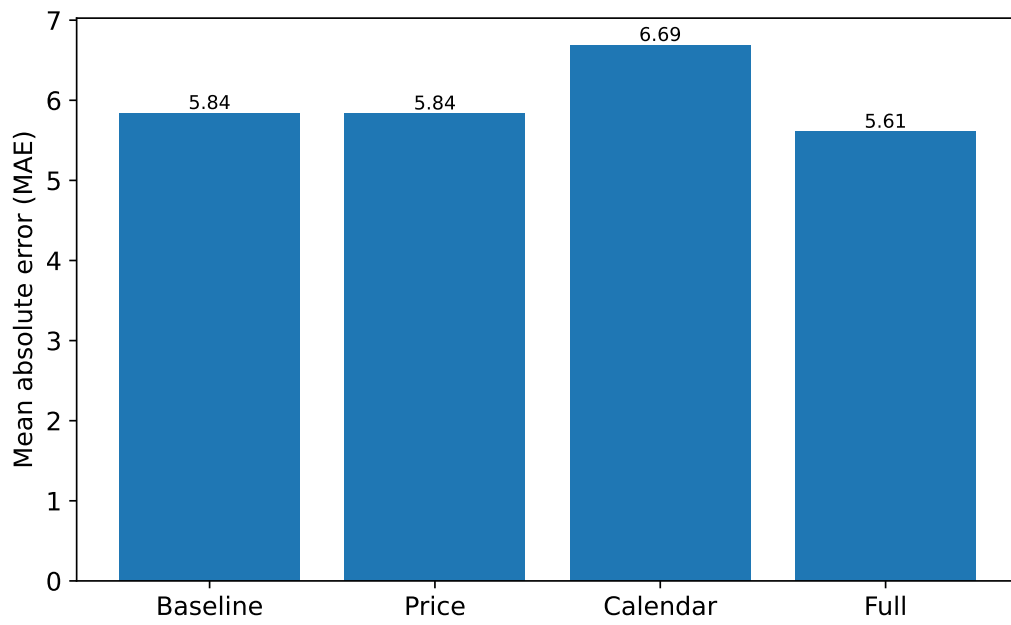


Figure 4: MAE for the four feature configurations evaluated in RQ1. The full multi-source configuration provides the lowest test-set MAE, while the calendar-enhanced configuration performs worse than the historical-demand baseline.

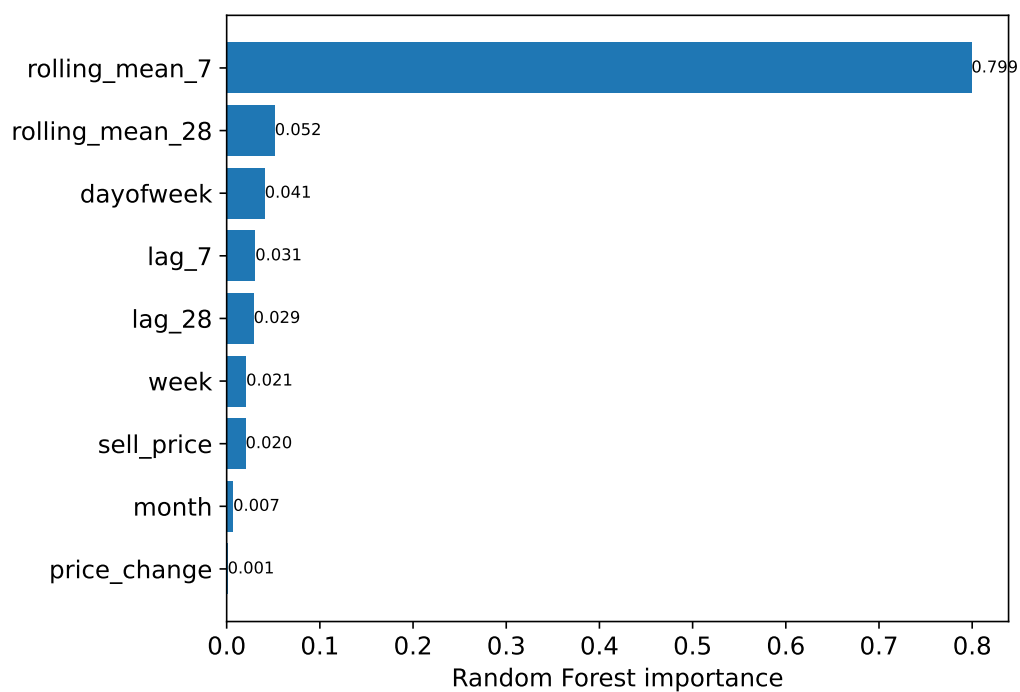


Figure 5: Random Forest feature-importance ranking for the engineered predictors. The 7-day rolling mean dominates the diagnostic, followed by the 28-day rolling mean, while individual pricing and calendar variables contribute smaller shares.

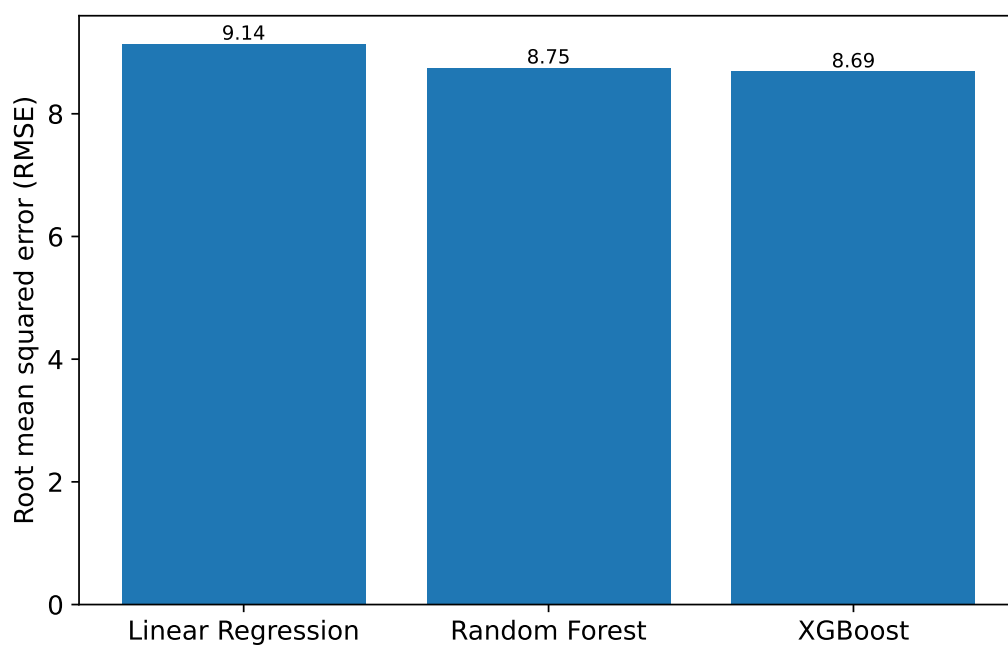


Figure 6: Core model comparison on the identical M5 chronological test set. XGBoost provides the lowest MAE and RMSE and the highest R^2 among Linear Regression, Random Forest, and XGBoost.

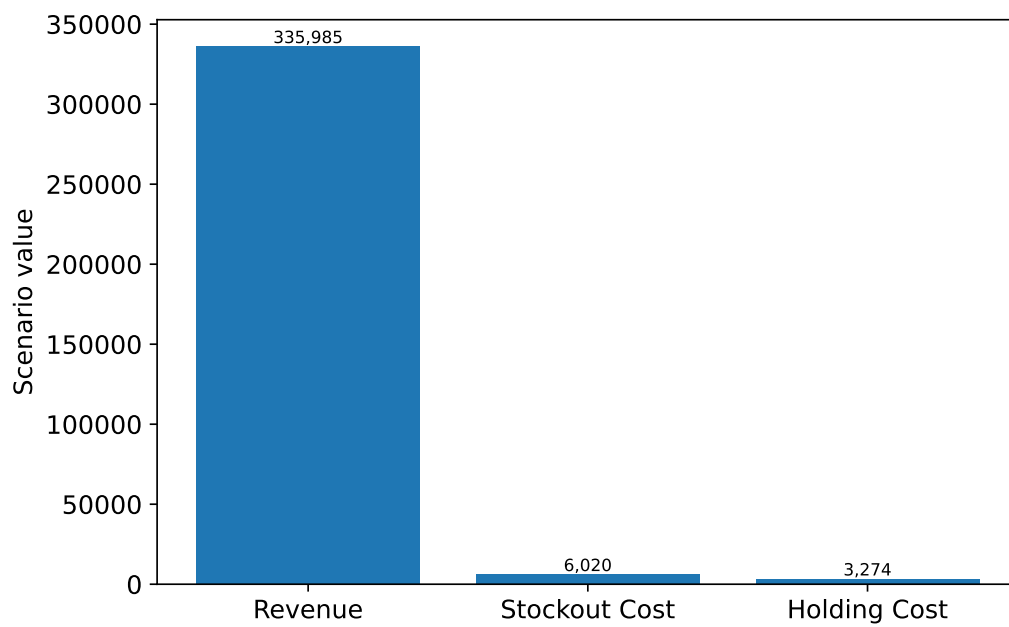


Figure 7: Financial quantities derived from LightGBM forecasts on the M5 test set. Revenue, stockout-cost exposure, and holding-cost exposure are scenario values calculated under the stated illustrative cost assumptions rather than observed retailer accounting outcomes.

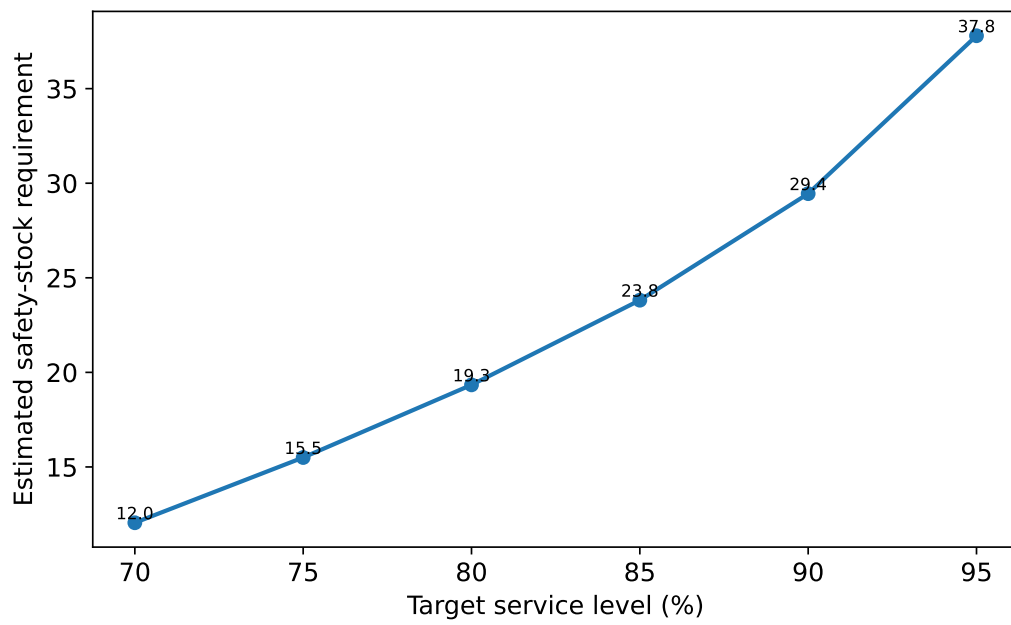


Figure 8: Estimated safety-stock requirement across target service levels from 70% to 95%. The estimates use the observed M5 LightGBM forecast-error dispersion and a seven-day lead-time assumption and are therefore scenario values.

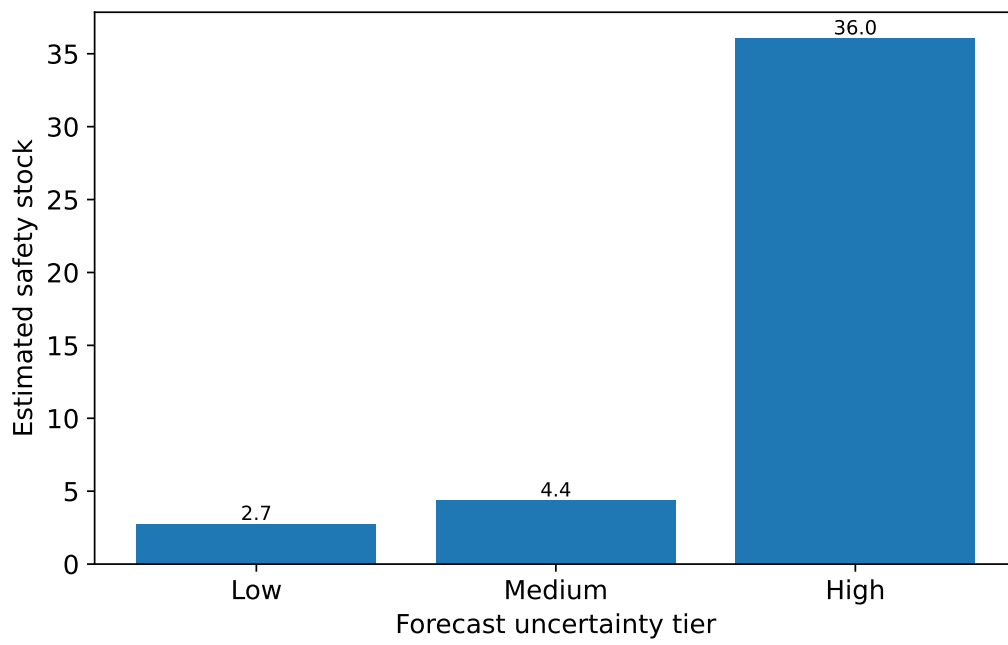


Figure 9: Forecast-error distribution and uncertainty-aware inventory diagnostics across the 14,000 M5 test observations. The high-error tail is associated with substantially larger estimated safety-stock requirements.

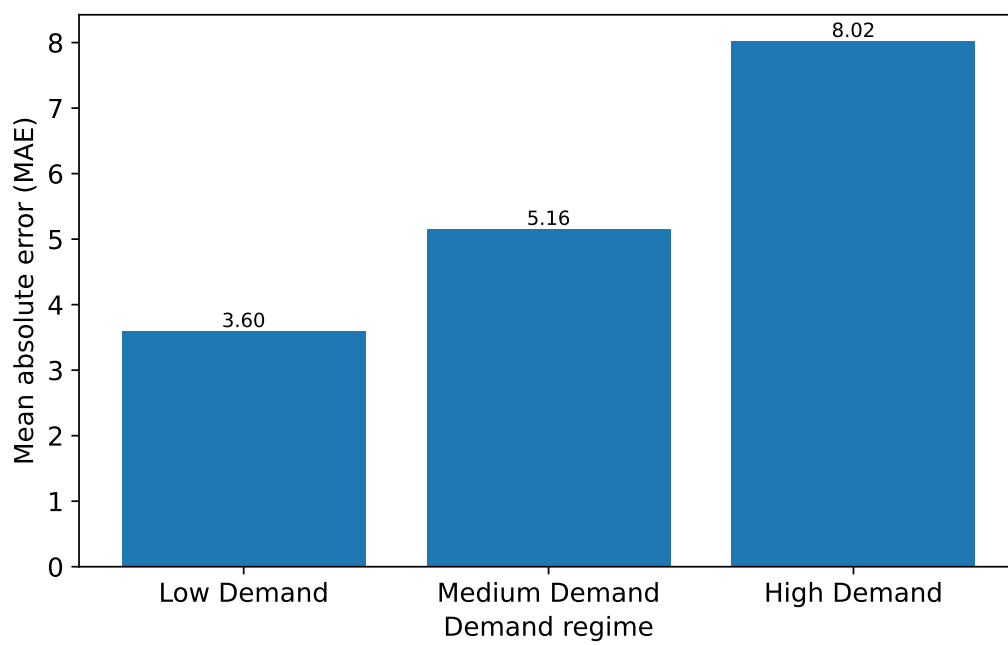


Figure 10: LightGBM MAE across low-, medium-, and high-demand regimes. Forecast error increases with demand intensity, with the high-demand regime showing the largest MAE.